Solution

ELECTRICITY -06

Class 10 - Science

1. **(a)** ammeter A_2 and voltmeter V_1

   Explanation: We should select instruments without any zero error.

2.
   (d) II

   Explanation: Voltmeter measures V across the parallel combination and the ammeter shows the current drawn by the resultant resistor.

3.
   (b) 8 Ω

   Explanation: 4Ω is short circuited. So 2 + 6 = 8 Ω

4.
   (c) ($i_A = i_B$); (V_A=V_B)

   Explanation: V is in parallel to the source. R_1 and R_2 are in parallel in A and series in B. So equivalent resistance is less and so current in A(i_A) is more than current in B(i_B).

5.
   (b) $R_1 > R_2 > R_3$

   Explanation:
   V has been plotted on the vertical axis, and I has been plotted on the horizontal axis
   $\Rightarrow$ R $\propto$ slope of V - I graph.
   Slope of graph R_1 is the highest and slope of graph R_3 is the least. Therefore, the value of resistance R_3 is the least. $R_1 > R_2 > R_3$

6.
   (d) A_1, A_2, A_3 and A_4

   Explanation: A_1 and A_2 show same reading. A_3 and A_4 connected to 2Ω resistor each, also carry equal current which is ($\frac{I}{2}$) [Current is divided equally here].

7.
   (d) 0 V in circuit A and 3V in circuit B.

   Explanation: Only circuit B, with a dot within the symbol of the plug key, is a closed circuit in which current is flowing and will show non-zero voltage. The voltmeter reading, for the set-ups shown, would be (nearly) equal to the voltage of the battery. ON/OFF state of the key is the only difference.

8.
   (b) Ammeter and rheostat

   Explanation: One common point only exists between ammeter, cell and rheostat.

9. **(a)** B and C

   Explanation: Polarities are reversed in V. R_1 and R_2 have only one common point.

10. **(a)** $V_A > V_B > V_C$ but $I_A > I_B > I_C$

    Explanation: The current at A is greater than that at the positions B and C. i.e. I_A> I_B> I_C. So $V_A$for A $V_A > A_B > V_C$.

11.
    (c) 0.5 amp

    Explanation: Given,
    I = 1 amp
    R = 5 ohm
    The potential difference across each resistor is the same as the resistors are connected in parallel and the voltage

across these resistors are the same.

Therefore, applied voltage, V = IR = 5V

Current through 4 ohm resistor = $\frac{V}{R} = \frac{5}{4}$ = 1.25 amp

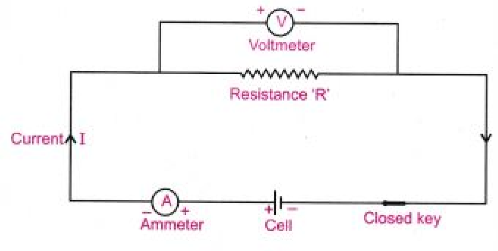
Current through 10 ohm resistor = $\frac{V}{R} = \frac{5}{10}$ = 0.5 amp

12. **(a)** 2.4 V

Explanation: 2.4 V

13. **(a)** a milliammeter, a resistor and a voltmeter

Explanation: Ammeter (X) in series and voltmeter (Z) in parallel to the resistor (Y).



14.

(c) voltmeter and connecting it across T and P with correct polarity

Explanation: Equivalent resistance in series can be found only when the p.d. across the series combination is known.

15.

(c) $i_1 < i_2$; $V_1 = V_2$

Explanation: The equivalent resistance, of a parallel combination of resistors, is less than the resistance of either of its two branches. The equivalent resistance, in circuit II, is therefore, less than ($R_1 + R_2$) (the equivalent resistance of circuit I) and hence the current flowing through it increases. The voltage reading, in both cases, is however, the same.

16.

(b) With S_2 open and S_1 closed A and B are lit

Explanation: With S_2 open and S_1 closed A and B are lit

17. **(a)** circuit A only

Explanation: In circuit B, the voltmeter is connected across R_1 only. So the potential across the combination is not measured.

18.

(b) 0 V in circuit I and 3V in circuit II

Explanation: Circuit I is an open circuit while circuit II is a closed circuit with resistors in parallel.

19.

(b) (-4 mA, +0.2 V) and (1 mA, 0.1 V) respectively

Explanation: Zero error → - 4mA, 0.2 volt

Least count = $\frac{10-0}{10} = \frac{10}{10} = 1\ mA\ ,\ \frac{1-0}{10} = \frac{1}{10} = 0.1\ volt$

20.

(d) incorrect reading for current I as well as potential difference V.

Explanation: Ammeter should not be shunted with voltmeter.

21. **(a)** 5 Ω

Explanation: V = 0.8 V, I = 160 mA = 160 × 10^{-3} A

$R = \frac{V}{I} = \frac{0.8}{160 \times 10^{-3}} = 5\Omega$

22. **(a)** $R_2 > R_1 > R_3$

Explanation: $R_2 > R_1 > R_3$

23.

(b) The Resistors R_1 and R_2 have been correctly connected in parallel.

Explanation: R_1 and R_2 has one common points.

24. **(a)** $9\,\Omega$

Explanation: In the given circuit, two $6\,\Omega$resistors are in a parallel combination, and this parallel combination is in series with another $6\,\Omega$ resistor.

For a parallel combination, the reciprocal of the effective resistance is equal to the sum of reciprocals of the individual resistances.

∴ The effective resistance of the parallel combination is given by**:** $\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2}$ or $\frac{1}{R_P} = \frac{1}{6} + \frac{1}{6}$ or $\frac{1}{R_P} = \frac{2}{6}$ or $R_P = 3\,\Omega$ The effective resistance of a series combination is equal to the sum of the individual resistances.

Therefore, the net effective resistance of the combination will be $R_S = 6\,\Omega + 3\,\Omega = 9\,\Omega$

25.

(b) 0 V in circuit I and 2 V in circuit II

Explanation: 0 V in circuit I as key open and 2V in circuit II.

26. **(a)** Student (A) will determine the equivalent resistance of the series combination while student (B) will determine the equivalent resistance of the parallel combination of the two resistors.

Explanation: R_1 and R_2 has one common point in (A) and two common points in (B).

27. **(a)** 6 amp

Explanation: Given,

$R_1 = 6\,\Omega$

$R_2 = 4\,\Omega$

V = 24V

From the given circuit diagram we know that the two resistances are connected in parallel

Therefore,

Current across $R_1 = I_1 = \frac{V}{R1} = \frac{24}{6}$ = 4 amp

Current across $R_2 = I_2 = \frac{V}{R2} = \frac{24}{4}$ = 6 amp

28.

(b) case X only

Explanation: The terminals are to be interchanged.

29.

(d) Student Z only

Explanation: Students Z has kept 1 common point not shared by any other device between R_1 and R_2 with the correct polarity of ammeter and voltmeter.

30. **(a)** A_1

Explanation: The current shown by A_1 is shared by others.

31.

(d) Both A and B

Explanation: Two common points should be there for R_1 and R_2 where the voltmeter is to be connected. It is not done.

32.

(d) Maximum in case (ii)

Explanation: In this case, two resistors are in series. Hence, their sum will be equal to their arithmetic sum. In the figure, the total resistance will be less than individual resistances because they are connected in parallel. A higher resistance produces more heat.

33. **(a)** 6A

Explanation: I = $\frac{v}{R_{eq}}$ = $\frac{8}{(4/3)}$ = 6 A

34. **(a)** Terminals of ammeter are wrongly connected.

Explanation: Terminals are to be interchanged.

35.

(d) 5.0 ohm

Explanation: $R = \frac{V}{I} = \frac{0.9\ volt}{180\ mA} = \frac{0.9 \times 10^3}{180} = \frac{900}{180} = 5\ \Omega$

36. **(a)** 9 Ω
Explanation: Again series and the parallel combination are used for the calculation of overall resistance of the given circuit.

37. **(a)** [(1, M), (2, N)], [(1, N), (2, P)] and [(1, M), (2, P)]
Explanation: The student has to find R_1, R_2 and (R_1 + R_2) for verification.

38.
(d) B
Explanation: Ammeter should be connected in series and the voltmeter in parallel along with resistor in parallel with correct polarities.

39.
(c) 6 A
Explanation: 6 A

40.
(c) is only circuit I
Explanation: The two resistors have 2 common points and are in parallel in both the circuits, but the voltage source is not connected properly in circuit II.

41.
(c) Correct reading for current I but incorrect reading for potential difference V.
Explanation: Ammeter is in series. But voltmeter is not in parallel. If ammeter has a significant resistance, voltmeter will also show deflection.

42.
(d) 0 V in case (i) and 3 V in case (ii)
Explanation: As the key plug is taken out, no current will flow in the circuit. So, reading of voltmeter will be zero in case (i). In case (ii), voltage across the parallel combination of 3 Ω and 2 Ω will remain 3 volt only.

43. **(a)** correct reading for current I, but incorrect reading for voltage V
Explanation: The voltmeter has to be put in parallel with the resistance being measured and not across the ammeter.

44.
(d) 1 V
Explanation: We know that the potential difference across a resistor is calculated as V = IR. Therefore, substituting the values we get potential difference across the resistor of 3Ω as 1V.

45. **(a)** (i)
Explanation: In this combination, positive terminal of next cell is adjacent to negative terminal of pervious cell.

46.
(c) I and III with III more reliable of the two
Explanation: Option with 4 and 2 is wrong as they are with voltmeters. Option I is nearest. The best option may be I and III with III more accuracy of the two since L.C. is less.

47.
(b) resistors have been connected correctly but the voltmeter has been wrongly connected
Explanation: Voltmeter should measure the potential difference across the combination of resistors. (in parallel/series)

48.
(c) circuit (I) only
Explanation: In II option, the deflection will be reverse (below zero).

49.
(c) Temperature T_2
Explanation:
Temperature T_2 is higher than T_1 since the slope ($\frac{\Delta I}{\Delta V}$) of the V - I graph at temperature T_2, and hence the reciprocal of the resistance, is smaller. Both the resistance and resistivity of a material vary with temperature. Resistivity of a conductor increases with increase in temperature.

50.

(d) 42 mA and 3.2 V

Explanation: The miliammeter and the voltmeter has zero error and least count. Milliammeter has negative error. Voltmeter has positive error.

51. **(a)** A_3, A_4

Explanation: A_3 and A_4 are in series so, the ammeters showing same reading.

52.

(c) 10

Explanation: $R = \frac{V}{I} = \frac{1.8V}{180\ mA} = \frac{1.8\times\ 10^3}{180} = \frac{1800}{180} = 10\ \Omega$

53. **(a)** $V_1 = V_2 = V_3$ and $I_1 \neq I_2 \neq I_3$

Explanation: R_1 and R_2 are in parallel and the voltmeter is connected in parallel to them in all the three cases. So $V_1 = V_2 = V_3$. The ammeter connected in series at different locations will indicate different current. $I_3 > I_1$ and I_2 without knowing values of R_1 and R_2 one cannot decide I_1 and I_2. If R_1 and R_2 given as same are taken as equal, then $I_1 = I_2$.

54.

(b) The same in all the cases

Explanation: The current record in the ammeter will be the same in all cases. None of the conditions change in any of the circuits.

55.

(d) $I_1 < I_2$ but $V_1 = V_2$

Explanation: R_{eq} in the two circuits are not same $R_{eq1} \geq R_{eq2}$ so $I_1 < I_2$. Voltmeter ends are connected to the cell in both the cases.

So $V_1 = V_2$

56.

(c) circuit (I) only

Explanation: Proper polarities in ammeter and voltmeter are connected. In II option, the deflection will be reverse (below zero).

57.

(d) 130 mA

Explanation: Least count = $\frac{500\ mA}{50} = 10\ mA$

No. of divisions = 13

$\therefore$ Reading = 130 mA

58.

(b) the two resistors have been connected correctly but both the voltmeter and the ammeter have not been connected correctly.

Explanation: Voltmeter is to be connected in parallel, while the ammeter should be connected in series.

59.

(b) 2.0 A

Explanation: 2.0 A

60.

(b) 12.83 V

Explanation: 12.83 V

61.

(b) 2

Explanation: The ammeter must be connected in series, between the battery and the series combination of the two resistors, and the voltmeter should be put in parallel across the series combination of the two resistors. All the polarities must also be correct.

62.

(d) M

Explanation: As J is shifted from K $\rightarrow$ M V increases with increasing current I.

63. **(a)** L

Explanation: Voltmeter in parallel and Ammeter in series to the series combination of the resistors with appropriate terminals is seen only in L.

64. **(a)** less than 3Ω

Explanation: Two points are common for the two resistors. So they are in parallel and the equivalent resistance should be less than the least 3Ω .

$\frac{1}{R_p} = \frac{1}{3} + \frac{1}{6} = \frac{2+1}{6} = \frac{1}{2}\Omega$

$\therefore$ R_p = 2Ω

65.

(d) IV

Explanation: "V" has to be connected in parallel and the potential of +ve terminal is greater than -ve terminal.

66.

(c) 0 V in circuit A and 3 V in circuit B

Explanation: Only circuit B, with a dot within the symbol of the plug key, is a closed circuit in which current is flowing and will show non-zero voltage. The voltmeter reading, for the set-ups shown, would be (nearly) equal to the voltage of the battery.

67. **(a)** (iv) in set up A and (i) in set up B

Explanation: With J at (iv) the potential difference is more in (A) and with J at (i) the potential difference is maximum and the equivalent resistance is least in (B).

68. **(a)** 9.9

Explanation: When the resistance arrangement from A is combined with the resistance arrangement B, the following is calculated:

$\frac{1}{R} = \frac{1}{10} + \frac{1}{1000}$

= $\frac{(100+1)}{1000}$

R = $\frac{1000}{101}$

R = 9.9 Ohm

Therefore, arrangement B has a lower combined resistance.

69.

(c) In the circuit (X) the resistors have been connected in parallel whereas these are connected in series in circuit (Y)

Explanation: There are two common points between them in X and one common point in Y.

70.

(c) R' is equivalent to the series combination of R_1 and R_2, R'' is equivalent to the parallel combination of R_1 and R_2

Explanation: I denotes series as one common point exists and II denotes parallel combination of R_1 and R_2 as 2 common points exists.

71. **(a)** Student B and C

Explanation: Resistors have one common point, voltmeter and ammeter are connected in proper way.

72.

(c) some reading in the ammeter but no reading in the voltmeter

Explanation: Key in series with the voltmeter is not closed. So it will not show any value.

73.

(d) all X, Y and Z cases

Explanation: Voltmeter has to be connected in parallel to the resistors where the potential drop has to be measured.

74.

(c) 9 Ω

Explanation: $R_S = R_1 + R_2$ = 3 + 6 = 9Ω as he has connected resistance R_1 and R_2 in series.

75.

(c) R_1, R_2 and V

Explanation: $R_1.R_2$ and V have two common points.